

NLP Project Progress Report

In-Context Learning: Chain-of-Thought vs Tree-of-Thoughts Prompting on Multi-Step Reasoning

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1 Project Title, Group Members, and Problem Definition

Title: In-Context Learning: Chain-of-Thought vs Tree-of-Thoughts Prompting on Multi-Step Reasoning (*no change since proposal*)

Task. Multi-step mathematical reasoning via prompting strategies applied to LLMs *without fine-tuning* (In-Context Learning, ICL). The NLP task is closed-form question answering: the model must produce a correct numeric answer after a chain of deductive steps.

Why it matters. LLMs frequently fail at multi-step problems when queried directly. CoT and ToT have demonstrated substantial improvements on such benchmarks; comparing them reveals how reasoning *structure* (linear vs. tree-shaped) affects accuracy.

- **Dataset:** GSM8K — grade-school math word problems requiring 2–8 reasoning steps.
- **Input:** Natural language problem (e.g., “*James has 3 apples and buys 4 more...*”).
- **Output:** Step-by-step reasoning chain followed by a final numeric answer.
- **Task type:** Question answering (closed-form, numeric) with multi-step reasoning.
- **Technical challenges:** (1) Error propagation across steps; (2) ToT requires 12–20 API calls per problem; (3) Evaluating intermediate reasoning quality; (4) Prompt sensitivity to few-shot example choice.

2 Dataset / Benchmark Status

Property	Details
Source	OpenAI; HuggingFace Hub: <code>openai/gsm8k</code> (config: <code>main</code>)
Total size	8,792 problems
Train / Val / Test	7,473 / ~747 (10% of train) / 1,319
Avg. length	~50 tokens; 2–8 reasoning steps
Answer format	#### <numeric>; regex: <code>r"####\s*([\d,\.]+)"</code>

Preprocessing: Downloaded via `datasets` library; answer normalization (strip commas, cast to float) implemented and tested. No text cleaning required. Tokenization is not applicable — prompting only, no model training. **Status: dataset downloaded and extraction script tested. ✓**

Limitations: Problems with 6–8 steps are prone to error propagation. ToT API cost is ~5–10× higher than CoT per problem. Dataset is English-only with a Western cultural context.

3 Literature Review Progress

Five foundational papers have been reviewed:

1. **Wei et al. (2022)** — *Chain-of-Thought Prompting Elicits Reasoning in LLMs* (NeurIPS). Introduced few-shot CoT; their 8-shot GSM8K prompt is adopted as Baseline 2.

2. **Kojima et al. (2022)** — *Large Language Models are Zero-Shot Reasoners* (NeurIPS). Showed that appending “Let’s think step by step.” elicits CoT without examples; forms Baseline 1.
3. **Yao et al. (2023)** — *Tree of Thoughts* (NeurIPS). Proposed BFS/DFS over reasoning branches with LLM-based evaluation; the main method in this work.
4. **Wang et al. (2022)** — *Self-Consistency Improves CoT Reasoning*. Majority voting over diverse CoT paths improves accuracy 10–20%; planned as an enhancement.
5. **Brown et al. (2020)** — *Language Models are Few-Shot Learners* (GPT-3, NeurIPS). Established the ICL paradigm; provides the theoretical backdrop for the entire study.

Technical direction: API-based prompting (no fine-tuning); two CoT baselines vs. ToT; GSM8K accuracy as the primary metric.

4 Baseline Models and Prompting Strategies

Baseline 1 — Zero-Shot Chain-of-Thought [✓ Implemented]

Prompt: Q: {problem} / A: Let’s think step by step. No examples; minimal cost; applicable to the full test set. **Result: 87.0%** accuracy on 100 GSM8K test samples with Llama 3.1 8B (Groq API).

Baseline 2 — Few-Shot Chain-of-Thought (8-shot) [✓ Implemented]

Eight worked examples from Wei et al. (2022) Appendix G prepended to the prompt. Directly comparable to ToT under identical model/test conditions. **Result: 85.0%** accuracy on 100 GSM8K test samples with Llama 3.1 8B (Groq API).

Main Method — Tree-of-Thoughts (BFS, branching=2, depth=2) [✓ Complete]

At each depth level, 2 candidate “thoughts” are generated (temperature=0.7) and scored *sure/maybe/impossible* by a separate evaluator prompt. Top-scoring branches are retained and expanded. **Result: 13.3%** accuracy on 30 GSM8K test samples with Llama 3.1 8B (Groq API). Each problem requires ~12 API calls; a 15-second inter-call sleep and 30-second inter-problem pause were required to stay within Groq free-tier rate limits.

5 Initial Experimental Results

Setup: Llama 3.1 8B (Groq API); 100 randomly sampled GSM8K test problems (seed=42); ToT uses 30 samples. Metric: accuracy = correct final answers / total problems evaluated.

Strategy	Accuracy	N	Status
Zero-Shot CoT	87.0%	100	✓ Complete
Few-Shot CoT (8-shot)	85.0%	100	✓ Complete
Tree-of-Thoughts (BFS-2)	13.3%	30	✓ Complete
Self-Consistency (CoT + vote)	—	—	Planned

Zero-shot CoT achieves 87.0% and few-shot CoT achieves 85.0% on the 100-sample evaluation set, while Tree-of-Thoughts achieves only **13.3%** on a 30-sample subset — a striking reversal of the results reported in Yao et al. (2023).

Zero-shot vs. Few-shot CoT: Contrary to typical expectations, few-shot CoT scores slightly below zero-shot CoT. This suggests Llama 3.1 8B has already internalised strong step-by-step reasoning from pre-training; the 8-shot demonstrations add minor noise rather than useful signal.

ToT underperformance: The poor ToT result is attributable to three compounding factors: (1) the evaluator prompt requires strong meta-cognitive reasoning that an 8B model cannot reliably perform, leading to incorrect branch pruning; (2) depth=2 is too shallow for multi-step math, causing the model to stop mid-reasoning and return an intermediate value as the final answer; (3) the original ToT results were obtained with GPT-4. These findings underscore that ToT is a large-model technique and does not transfer to smaller models without significant adaptation.

6 Planned Improvements and Technical Direction

1. **Self-Consistency + Few-Shot CoT:** Run few-shot CoT 10 times (temperature=0.7), select by majority vote (Wang et al., 2022). Expected +8–15 pp over standard few-shot CoT.
2. **ToT depth/branching ablation:** Sweep branching $\in \{1, 2, 3, 5\}$, depth $\in \{2, 4, 6\}$; plot cost-vs-accuracy Pareto frontier. Branching=1 degenerates to greedy CoT.
3. **Larger-model comparison:** Repeat key experiments on 200-sample subset using a larger model (e.g., Llama 3.3 70B or GPT-4o). Expected accuracy gain over Llama 3.1 8B on both CoT and ToT.
4. **Prompt sensitivity:** Run few-shot CoT with 3 independent random 8-example sets; measure variance attributable to example choice.
5. **Error analysis:** Manual inspection of incorrect predictions; categorize error types (Section 8).

7 Ablation and Prompt Sensitivity Plan

- **Few-shot example selection:** Same 8-shot strategy run with 3 independently sampled example sets; accuracy variance quantifies sensitivity to example choice.
- **Zero-shot vs. few-shot CoT:** Direct comparison already underway; establishes marginal value of demonstrations.
- **ToT branching factor:** BFS branching $\in \{1, 2, 3, 5\}$; branching=1 reduces ToT to greedy CoT. Cost-accuracy trade-off table will guide budget-optimal configuration.
- **Decoding temperature:** Greedy (temperature=0) vs. stochastic (0.7) for both few-shot CoT and ToT thought generation.
- **Problem difficulty sub-groups:** GSM8K partitioned into short (≤ 3 steps) and long (≥ 6 steps) subsets; method accuracy reported separately to test whether ToT’s advantage concentrates in harder problems.

8 Error Analysis and Generation Quality Analysis

Error categories: Each incorrect prediction is classified as (1) Arithmetic error, (2) Misinterpretation of the problem, (3) Step omission, or (4) Early termination.

Qualitative protocol: 10 randomly sampled incorrect predictions per strategy inspected manually; annotated with (a) first error step, (b) whether the error is “fluent but wrong”, (c) whether a subsequent step could recover.

Example annotation:

Problem: “A store has 48 apples, sells 5/hour, after 6 h a delivery doubles the stock. How many?”
Gold: 66 Model output: 72 Error type: Arithmetic (incorrect final addition) CoT trace: 18
 $\times 2 = 36 \dots 36 + 36 \neq 66$

ToT-specific: Evaluator logs inspected to identify cases where a demonstrably wrong thought receives a *sure* rating (calibration failure). Instances of fluent but mathematically incorrect output counted separately to distinguish language quality from reasoning reliability.

9 Ethical Considerations and Bias

- **Hallucination risk:** CoT and ToT can produce linguistically plausible but mathematically incorrect reasoning chains. High-stakes applications (financial, medical) may over-trust fluent-but-wrong outputs.
- **Evaluation limitations:** Exact numeric-match accuracy ignores reasoning quality; LLM-as-judge evaluation of intermediate steps is being considered as a complementary metric.
- **Sampling bias:** ToT evaluation is limited to 30 samples due to high per-problem API call count on free-tier infrastructure. All reported accuracies include sample size for transparency.
- **Few-shot example bias:** Unrepresentative 8-shot examples may inflate reported accuracy. The prompt sensitivity ablation (Section 7) directly tests this risk.
- **Dataset bias:** GSM8K is English-only with American cultural context; conclusions should not be generalized to other languages or cultural settings.

10 GitHub and Reproducibility Status

Repository: https://github.com/OguzhanCKL/CENG467_Project1

Reproducibility Item	Status
README with setup & usage	✓
requirements.txt with pinned versions	✓
.env.example for API key management	✓
.env excluded from version control	✓
Random seed fixed (seed=42 in all sampling)	✓
Data download + answer extraction script	✓
Unified experiment runner (src/run_experiments.py)	✓
Results saved as JSON for post-hoc analysis	✓

Run an experiment:

```
pip install -r requirements.txt
cp .env.example .env # add GROQ_API_KEY
python data/download_data.py
python src/run_experiments.py --strategy few_shot_cot --n_samples 100
```

11 Current Challenges and Next Steps

Current challenges:

1. **API rate limits (ToT) — Resolved:** Free-tier Groq API rate limits were mitigated with a 15-second inter-call sleep and 30-second inter-problem pause. The 30-sample ToT experiment completed successfully.
2. **ToT evaluator calibration — Key finding:** The evaluator prompt performs very poorly with Llama 3.1 8B (8B parameters), frequently assigning *sure* to incorrect steps. This is a fundamental small-model limitation and the central finding of the project, explaining the 13.3% ToT accuracy.
3. **Reproducibility at temperature > 0:** Final comparisons use temperature=0; temperature=0.7 reserved for self-consistency and ToT thought generation only.
4. **Time constraints:** Full ablation grid (~50 conditions); prioritization applied to highest-value comparisons.

Remaining timeline:

Task	Target
ToT experiment — <i>complete</i>	✓ Done
Implement Self-Consistency baseline	Week +1
Ablation study (branching, temperature)	Weeks +2–3
Error analysis (manual annotation)	Weeks +3–4
Larger-model comparison (200-sample subset)	Week +3
Write final report	Week +5

Report generated: May 2026. All three strategies evaluated; full ablation and error analysis in progress.